Laboratory Course: [BCS-352] IT Infrastructure Management Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Prof. Amit Saxena (Dept. of**

Official Laboratory Manual: Linux System Administration & Virtualization

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1. Course Overview & Laboratory Objectives

- Focus: Hands-on Linux systems administration, VM configuration, and service deployment.
- Environment: Ubuntu Server / CentOS virtualized on Type-1 and Type-2 hypervisors.
- Competencies: Shell scripting, user permissions, networking setup, and web server configuration.

2. Key Prescribed Practical Exercises

- Lab 1-4: Linux terminal mastery, user permissions (chmod 755/644), chown, and sudoers configuration.
- Lab 5-7: Hypervisor setup: creating and configuring VM snapshots and NAT/Bridged virtual networks.
- Lab 8-10: Text stream processing with vi editor, grep, awk, sed, and diff patch generation.
- Lab 11-12: Enterprise service deployment: Apache web server, SSH key pairs, and firewall (ufw).